

《数据库系统原理》实验报告（4）					
题目：SQL 综合实验					
学号		姓名		日期	2025.4.30
实验环境： Docker-desktop 4.40.0 oceanbase-ce 4.3.5.1					
实验步骤及结果截图： (1)建立数据库，在数据库中建表，表内字段的类型可以自行定义（合理即可），注意建表时不要忽略各表的主键约束和表间的外键约束 <pre>CREATE TABLE Movie (Movie_no VARCHAR(10), Movie_name VARCHAR(50), Director VARCHAR(30), Rating DECIMAL(3,1), End_date DATETIME, PRIMARY KEY (Movie_no));</pre> <pre>obclient(root@sys)[exp4]> CREATE TABLE Viewer (-> Viewer_no VARCHAR(10), -> Viewer_name VARCHAR(30), -> Age INT, -> PRIMARY KEY (Viewer_no) ->); Query OK, 0 rows affected (0.083 sec) obclient(root@sys)[exp4]> desc Viewer; +-----+-----+-----+-----+-----+-----+ Field Type Null Key Default Extra +-----+-----+-----+-----+-----+-----+ Viewer_no varchar(10) NO PRI NULL Viewer_name varchar(30) YES NULL Age int(11) YES NULL +-----+-----+-----+-----+-----+-----+ 3 rows in set (0.014 sec)</pre> <pre>CREATE TABLE Viewer (Viewer_no VARCHAR(10), Viewer_name VARCHAR(30), Age INT, PRIMARY KEY (Viewer_no));</pre>					

```
obclient(root@sys)[exp4]> CREATE TABLE Movie (
->   Movie_no VARCHAR(10),
->   Movie_name VARCHAR(50),
->   Director VARCHAR(30),
->   Rating DECIMAL(3,1),
->   End_date DATETIME,
->   PRIMARY KEY (Movie_no)
-> );
Query OK, 0 rows affected (0.098 sec)

obclient(root@sys)[exp4]> desc Movie;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Movie_no | varchar(10) | NO | PRI | NULL | |
| Movie_name | varchar(50) | YES | | NULL | |
| Director | varchar(30) | YES | | NULL | |
| Rating | decimal(3,1) | YES | | NULL | |
| End_date | datetime | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.014 sec)
```

```
CREATE TABLE Watch (
  S_no VARCHAR(10),
  Viewer_no VARCHAR(10),
  Movie_no VARCHAR(10),
  Watch_date DATETIME,
  PRIMARY KEY (S_no, Viewer_no, Movie_no),
  FOREIGN KEY (Viewer_no) REFERENCES Viewer(Viewer_no) ON DELETE CASCADE,
  FOREIGN KEY (Movie_no) REFERENCES Movie(Movie_no) ON DELETE CASCADE
);
```

```
obclient(root@sys)[exp4]> CREATE TABLE Watch (
->   S_no VARCHAR(10),
->   Viewer_no VARCHAR(10),
->   Movie_no VARCHAR(10),
->   Watch_date DATETIME,
->   PRIMARY KEY (S_no, Viewer_no, Movie_no),
->   FOREIGN KEY (Viewer_no) REFERENCES Viewer(Viewer_no) ON DELETE CASCADE,
->   FOREIGN KEY (Movie_no) REFERENCES Movie(Movie_no) ON DELETE CASCADE
-> );
Query OK, 0 rows affected (0.115 sec)

obclient(root@sys)[exp4]> desc Watch;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| S_no | varchar(10) | NO | PRI | NULL | |
| Viewer_no | varchar(10) | NO | PRI | NULL | |
| Movie_no | varchar(10) | NO | PRI | NULL | |
| Watch_date | datetime | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.015 sec)
```

(2)插入样例数据

insert into Movie values

('M001', '星际穿越', '克里斯托弗·诺兰', 9.3, '2024-05-01'),

('M002', '泰坦尼克号', '詹姆斯·卡梅隆', 9.1, '2024-04-10'),

('M003', '盗梦空间', '克里斯托弗·诺兰', 8.8, '2024-04-20'),

('M004', '科幻冒险之旅', '张三', 7.5, '2024-04-18'),

('M005', '爱情故事', '李四', 7.0, '2024-04-25');

insert into Viewer values

('V001', '李明', 25),

('V002', '王红', 30),

```
( 'V003', '张磊', 22),
( 'V004', '赵颖', 28),
( 'V005', '孙阳', 35);

insert into Watch values
( '1', 'V001', 'M001', '2024-03-15'),
( '2', 'V001', 'M001', '2024-03-20'),
( '2', 'V001', 'M002', '2024-03-20'),
( '3', 'V002', 'M002', '2024-03-25'),
( '1', 'V002', 'M003', '2024-04-01'),
( '2', 'V003', 'M001', '2024-04-05'),
( '2', 'V004', 'M002', '2024-04-12'),
( '1', 'V005', 'M003', '2024-04-14');
```

```
obclient(root@sys)[exp4]> insert into Movie values
-> ( 'M001', '星际穿越', '克里斯托弗·诺兰', 9.3, '2024-05-01'),
-> ( 'M002', '泰坦尼克号', '詹姆斯·卡梅隆', 9.1, '2024-04-10'),
-> ( 'M003', '盗梦空间', '克里斯托弗·诺兰', 8.8, '2024-04-20'),
-> ( 'M004', '科幻冒险之旅', '张三', 7.5, '2024-04-18'),
-> ( 'M005', '爱情故事', '李四', 7.0, '2024-04-25');
Query OK, 5 rows affected (0.016 sec)
Records: 5 Duplicates: 0 Warnings: 0

obclient(root@sys)[exp4]> insert into Viewer values
-> ( 'V001', '李明', 25),
-> ( 'V002', '王红', 30),
-> ( 'V003', '张磊', 22),
-> ( 'V004', '赵颖', 28),
-> ( 'V005', '孙阳', 35);
Query OK, 5 rows affected (0.018 sec)
Records: 5 Duplicates: 0 Warnings: 0

obclient(root@sys)[exp4]> insert into Watch values
-> ( '1', 'V001', 'M001', '2024-03-15'),
-> ( '2', 'V001', 'M001', '2024-03-20'),
-> ( '2', 'V001', 'M002', '2024-03-20'),
-> ( '3', 'V002', 'M002', '2024-03-25'),
-> ( '1', 'V002', 'M003', '2024-04-01'),
-> ( '2', 'V003', 'M001', '2024-04-05'),
-> ( '2', 'V004', 'M002', '2024-04-12'),
-> ( '1', 'V005', 'M003', '2024-04-14');
Query OK, 8 rows affected (0.006 sec)
Records: 8 Duplicates: 0 Warnings: 0
```

(3)查询电影名称中包含“科幻”的电影信息，输出所有信息（包括电影名称、电影编号、导演、评分、电影停映日期），并按照评分降序排列

```
SELECT Movie_name, Movie_no, Director, Rating, End_date
FROM Movie
WHERE Movie_name LIKE '%科幻%'
ORDER BY Rating DESC;
```

```
obclient(root@sys)[exp4]> SELECT Movie_name, Movie_no, Director, Rating, End_date
-> FROM Movie
-> WHERE Movie_name LIKE '%科幻%'
-> ORDER BY Rating DESC;

+-----+-----+-----+-----+-----+
| Movie_name | Movie_no | Director | Rating | End_date |
+-----+-----+-----+-----+-----+
| 科幻冒险之旅 | M004 | 张三 | 7.5 | 2024-04-18 00:00:00 |
+-----+-----+-----+-----+-----+
1 row in set (0.004 sec)
```

(4)查询观看了电影名为“泰坦尼克号”的观众信息，输出该观众的编号、姓名和年龄，并按照观众编号升序排列

```
SELECT DISTINCT v.Viewer_no, v.Viewer_name, v.Age
FROM Viewer v
JOIN Watch w ON v.Viewer_no = w.Viewer_no
JOIN Movie m ON w.Movie_no = m.Movie_no
WHERE m.Movie_name = '泰坦尼克号'
ORDER BY v.Viewer_no ASC;
```

```
obclient(root@sys)[exp4]> SELECT DISTINCT v.Viewer_no, v.Viewer_name, v.Age
-> FROM Viewer v
-> JOIN Watch w ON v.Viewer_no = w.Viewer_no
-> JOIN Movie m ON w.Movie_no = m.Movie_no
-> WHERE m.Movie_name = '泰坦尼克号'
-> ORDER BY v.Viewer_no ASC;

+-----+-----+-----+
| Viewer_no | Viewer_name | Age |
+-----+-----+-----+
| V001 | 李明 | 25 |
| V002 | 王红 | 30 |
| V004 | 赵颖 | 28 |
+-----+-----+-----+
3 rows in set (0.020 sec)
```

(5)统计每个观众的观影信息，输出每个观众的编号、观看的电影名称和观看日期

```
SELECT v.Viewer_no, m.Movie_name, w.Watch_date
FROM Viewer v
JOIN Watch w ON v.Viewer_no = w.Viewer_no
JOIN Movie m ON w.Movie_no = m.Movie_no
ORDER BY v.Viewer_no, w.Watch_date;
```

```
obclient(root@sys)[exp4]> SELECT v.Viewer_no, m.Movie_name, w.Watch_date
-> FROM Viewer v
-> JOIN Watch w ON v.Viewer_no = w.Viewer_no
-> JOIN Movie m ON w.Movie_no = m.Movie_no
-> ORDER BY v.Viewer_no, w.Watch_date;
```

Viewer_no	Movie_name	Watch_date
V001	星际穿越	2024-03-15 00:00:00
V001	星际穿越	2024-03-20 00:00:00
V001	泰坦尼克号	2024-03-20 00:00:00
V002	泰坦尼克号	2024-03-25 00:00:00
V002	盗梦空间	2024-04-01 00:00:00
V003	星际穿越	2024-04-05 00:00:00
V004	泰坦尼克号	2024-04-12 00:00:00
V005	盗梦空间	2024-04-14 00:00:00

8 rows in set (0.003 sec)

(6)查询所有已停映电影的信息，输出观众编号、姓名、电影名称和观看日期，并按观看日期降序排列

P.S.已停映电影指的是“现实日期”大于电影停映日期字段的电影，“现实日期”以4月15日为例。

```
SELECT v.Viewer_no, v.Viewer_name, m.Movie_name, w.Watch_date
FROM Viewer v
JOIN Watch w ON v.Viewer_no = w.Viewer_no
JOIN Movie m ON w.Movie_no = m.Movie_no
WHERE m.End_date < '2025-04-15'
ORDER BY w.Watch_date DESC;
```

```
obclient(root@sys)[exp4]> SELECT v.Viewer_no, v.Viewer_name, m.Movie_name, w.Watch_date
-> FROM Viewer v
-> JOIN Watch w ON v.Viewer_no = w.Viewer_no
-> JOIN Movie m ON w.Movie_no = m.Movie_no
-> WHERE m.End_date < '2025-04-15'
-> ORDER BY w.Watch_date DESC;
```

Viewer_no	Viewer_name	Movie_name	Watch_date
V005	孙阳	盗梦空间	2024-04-14 00:00:00
V004	赵颖	泰坦尼克号	2024-04-12 00:00:00
V003	张磊	星际穿越	2024-04-05 00:00:00
V002	王红	盗梦空间	2024-04-01 00:00:00
V002	王红	泰坦尼克号	2024-03-25 00:00:00
V001	李明	星际穿越	2024-03-20 00:00:00
V001	李明	泰坦尼克号	2024-03-20 00:00:00
V001	李明	星际穿越	2024-03-15 00:00:00

8 rows in set (0.009 sec)

(7)查询观看了“星际穿越”但没有观看“盗梦空间”的观众信息，输出这些观众的编号，并按照编号升序排列。

```
SELECT DISTINCT w1.Viewer_no
FROM Watch w1
JOIN Movie m1 ON w1.Movie_no = m1.Movie_no
LEFT JOIN (
    SELECT w2.Viewer_no
    FROM Watch w2
    JOIN Movie m2 ON w2.Movie_no = m2.Movie_no
```

```
WHERE m2.Movie_name = '盗梦空间'
) w2 ON w1.Viewer_no = w2.Viewer_no
WHERE m1.Movie_name = '星际穿越'
AND w2.Viewer_no IS NULL
ORDER BY w1.Viewer_no ASC;
```

```
obclient(root@sys)[exp4]> SELECT DISTINCT w1.Viewer_no
-> FROM Watch w1
-> JOIN Movie m1 ON w1.Movie_no = m1.Movie_no
-> LEFT JOIN (
->   SELECT w2.Viewer_no
->   FROM Watch w2
->   JOIN Movie m2 ON w2.Movie_no = m2.Movie_no
->   WHERE m2.Movie_name = '盗梦空间'
-> ) w2 ON w1.Viewer_no = w2.Viewer_no
-> WHERE m1.Movie_name = '星际穿越'
-> AND w2.Viewer_no IS NULL
-> ORDER BY w1.Viewer_no ASC;

+-----+
| Viewer_no |
+-----+
| V001      |
| V003      |
+-----+
2 rows in set (0.011 sec)
```

(8) 创建一个过程，使之能够实现如下功能

修改观影表，增加字段“重复观看状态”（字段名为“Repeat_state”），字段含义为表示某观众是否多次观看某电影；

并根据表中已有数据为该字段赋值（所赋的值与表定义时的数据类型保持一致即可，比如可以定义多次观看某电影的“重复观看状态”为 True，只看过一次某电影的“重复观看状态”为 False），要求使用 if 语句进行条件判断。

```
DELIMITER $$
CREATE PROCEDURE update_repeat_state()
BEGIN
    -- 1. 修改 Watch 表，增加 Repeat_state 字段，类型为 BOOLEAN
    ALTER TABLE Watch
    ADD COLUMN Repeat_state BOOLEAN DEFAULT FALSE;

    -- 2. 更新 Repeat_state 字段
    -- 使用子查询和 IF 语句判断每个 Viewer_no 和 Movie_no 组合的观看次数
    UPDATE Watch w
    SET Repeat_state = (
        SELECT IF(COUNT(*) > 1, TRUE, FALSE)
        FROM Watch w2
        WHERE w2.Viewer_no = w.Viewer_no
        AND w2.Movie_no = w.Movie_no
    );
END$$
DELIMITER ;
```

```
obclient(root@sys)[exp4]> CREATE UNIQUE INDEX Movie_name_index
-> ON Movie (Movie_name DESC);
Query OK, 0 rows affected (0.386 sec)
```

obclient(root@sys)[exp4]> SHOW INDEX FROM Movie;

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	Index_type	Comment	Index_comment	Visible	Expression
Movie	0	PRIMARY	1	Movie_no	A	NULL	NULL	NULL		BTREE	available		YES	NULL
Movie	0	Movie_name_index	1	Movie_name	A	NULL	NULL	NULL	YES	BTREE	available		YES	NULL

12 rows in set (0.030 sec)

出现的问题:

使用过程增加字段“重复观看状态”时失败

在调用 update_repeat_state() 过程时, SQL 语句出现列名重复的问题, 显示 repeat_state 这个列名可能被重复引用了, 无法正常运行。

```
obclient(root@sys)[exp4]> CALL update_repeat_state();
ERROR 1060 (42S21): Duplicate column name 'Repeat_state'
[172.17.0.2:2882] [2025-04-30 15:39:10.444253] [YB42AC110002-00063400B7472744-0-0]
```

过程重复执行问题

如果多次调用 update_repeat_state(), 而没有检查 Repeat_state 列是否存在, 会导致列名重复的错误

解决方案:

使用过程增加字段“重复观看状态”时失败

经检查, Watch 表中已经存在 repeat_state 列, 可能是因为之前创建的错误的过程, 虽然没有全部正确运行, 但是加上了这一列, 删去着一列后重新运行过程, 即可成功。

```
obclient(root@sys)[exp4]> desc Watch;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| S_no       | varchar(10) | NO   | PRI | NULL    |       |
| Viewer_no  | varchar(10) | NO   | PRI | NULL    |       |
| Movie_no   | varchar(10) | NO   | PRI | NULL    |       |
| Watch_date | datetime   | YES  |     | NULL    |       |
| Repeat_state | tinyint(1) | YES  |     | 0       |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.002 sec)
```

过程重复执行问题的解决

修改后的存储过程, 添加列存在性检查。

```
IF column_exists = 0 THEN
    ALTER TABLE Watch
    ADD COLUMN Repeat_state BOOLEAN DEFAULT FALSE;
END IF;
```